

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 50%

1. Grey Level Transformation

- (a) Explain the purpose of grey level transformation in image enhancement.
- (b) Apply a negative transformation $s = 255 - r$ to the pixel values:

$$r = [50, 100, 150, 200]$$

Compute the transformed values.

2. Log Transformation

- (a) Explain how log transformation enhances low-intensity pixels.
- (b) Given $s = c \log(1 + r)$, where $c = 10$, compute s for:

$$r = [0, 10, 50, 100]$$

3. Histogram Processing

- (a) Explain the role of histogram equalization in contrast enhancement.
- (b) Given a 3-bit image with histogram:

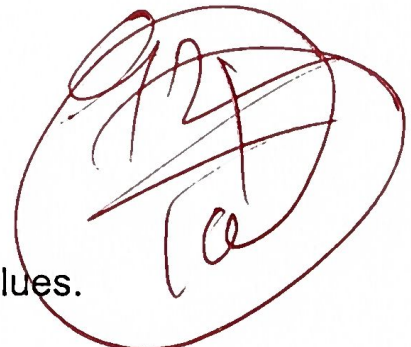
Intensity Frequency

0	2
1	2
2	4
3	8

Compute the cumulative distribution and equalized intensity values.

4. Spatial Filtering (Smoothing)

- (a) Explain the purpose of spatial filtering for smoothing.
- (b) Apply a 3×3 averaging filter to the following region:



[10 20 30 20 30 40 30 40 50]

Compute the new center pixel value.

5. Spatial Filtering (Sharpening)

(a) Explain how sharpening enhances image details.

(b) Apply the Laplacian mask:

$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$

to the matrix:

[10 10 10, 10 20 10, 10 10 10]

Compute the output at the center.

6. Frequency Domain Filtering

(a) Differentiate low-pass and high-pass filtering in frequency domain.

(b) Given frequency components:

$F = [100, 80, 40, 20]$

Apply a low-pass filter that retains only the first two components. Write the filtered output.

7. Thresholding

(a) Explain global thresholding in segmentation.

(b) Given pixel values:

[200]

Apply threshold $T = 100$. Assign values (0 or 1).

8. Edge Detection

(a) Explain the concept of edge detection.

(b) Apply the Sobel operator (horizontal):

$\begin{bmatrix} -1 & -2 & -1 & 0 & 0 & 0 & 1 & 2 & 1 \end{bmatrix}$

to:

[10 10 10, 20 20 20, 30 30 30]

Compute the gradient at the center.

9. Region-Based Segmentation

(a) Explain region growing in segmentation.

(b) Given pixel intensities:

[1 0 2]

If seed = 50 and threshold difference = 5, identify pixels belonging to the region.

10. Combined Enhancement & Segmentation

(a) Explain why smoothing is applied before segmentation.

(b) Given image values:

[1 0 5]

First apply averaging (assume mean = 48), then threshold at $T = 50$.

Determine final segmented output.

11. Power-Law (Gamma) Transformation

A grayscale image is enhanced using power-law transformation.

(a) Interpret the role of gamma correction in image enhancement.

(b) Justify the use of transformation $s = cr^\gamma$, where $\gamma = 0.5$, $c = 1$.

(c) Apply the transformation step-by-step to pixel values:

$r = [25, 100, 225]$.

(d) Compute the transformed values accurately.

(e) Interpret the results in terms of brightness enhancement.

12. Contrast Stretching

An image has intensity values in the range [50, 150].

(a) Interpret the need for contrast stretching.

(b) Justify linear transformation to range [0, 255].

(c) Apply transformation step-by-step for values:

$r = [50, 100, 150]$.

(d) Compute the output values accurately.

(e) Interpret improvement in contrast.

13. Histogram Equalization (4-bit Image)

A 4-bit image has the following histogram:

Intensity Frequency

0	4
1	6
2	10
3	6

(a) Interpret the need for equalization.

(b) Justify use of CDF.

(c) Compute cumulative distribution.

(d) Compute new intensity levels.

(e) Interpret contrast enhancement.

14. Median Filtering

A noisy image contains salt-and-pepper noise.

(a) Interpret noise characteristics.

(b) Justify median filter usage.

(c) Apply filter to:

10 200 10, 200 50 200, 10 200 10

(d) Compute center value.

(e) Interpret noise removal effectiveness.

15. Gaussian Smoothing

An image is smoothed using Gaussian filter.

(a) Interpret smoothing objective.

(b) Justify Gaussian filter over averaging.

(c) Apply filter:

[1 2 1, 2 4 2, 1 2 1]

to matrix:

[10 20 10, 20 40 20, 10 20 10]

(d) Compute center pixel.

(e) Interpret smoothing effect.

Code of Conduct:

I Vignesh / 192525013 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Vignesh

Q1. Grey level transformation

(a) Grey level transformation is used to modify pixel intensity values to improve image quality

(b) Negative transformation

Given : $S = 255 - x$

x	Calculation	S
50	$255 - 50$	205
100	$255 - 100$	155
150	$255 - 150$	105
200	$255 - 200$	55

Final output : [205, 155, 105, 55]

Q2. (a) log transformation

log transformation enhances low-intensity pixels while compressing high-intensity values. This makes details in dark regions of an image more visible.

(b) Calculation

x	Calculation	S
0	$10 \log(1)$	0.00
10	$10 (\log 11)$	10.41
50	$10 \log(51)$	17.08
100	$10 \log(101)$	20.04

Final output : [0.00 , 10.41 , 17.08 , 20.04]

Q3. Histogram equalization

Histogram equalization improves contrast by redistributing pixel intensities uniformly. It enhances the visibility of details, especially in low-contrast image.

(b) Calculation

Intensity	frequency	cumulative	equalized
0	2	2	0
1	2	4	0
2	4	8	1
3	8	15	3

Q4. Spatial filtering

Spatial Smoothing reduces noise and small variations in an image by averaging neighboring pixel values. It produces a smoother image & improves preprocessing for segmentation.

(b) Calculation

$$\begin{bmatrix} 10 & 20 & 30 \\ 20 & 30 & 40 \\ 30 & 40 & 50 \end{bmatrix}$$

$$\text{Average} \therefore \frac{10 + 20 + 30 + 20 + 30 + 40 + 30 + 40 + 50}{9} = \frac{270}{9} = 30$$

New center pixel = 30

Q5. Spatial filtering (Sharpening)

Sharpening enhances image details by highlighting edges & fine features. It increases the contrast between neighboring pixels.

(b) Calculation

Laplacian mask:

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Image matrix

$$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 20 & 10 \\ 10 & 10 & 10 \end{bmatrix}$$

center pixel

$$(4 \times 20) - (10 + 10 + 10 + 10) = 80 - 40 = 40$$

output at center = 40

69. Frequency Domain filtering

frequency domain filtering processes an image in the frequency domain. A low-pass filter retains low-frequency components and removes high-frequency noise, while a high-pass filter retains high-frequency.

(b) Calculation

Given frequency components:

$$f = [100, 80, 40, 20]$$

Apply a low-pass filter that retains only the first two components

$$\text{filtered output} : [100, 80, 0, 0]$$

70. Global thresholding

Global thresholding is a segmentation technique that separates an image into foreground & background using a single

pixels above the threshold are assigned 1, and those below it are assigned 0

(b) Calculation

Given pixel value : 200

Threshold : $T = 100$

Since $200 > 100$, assign 1

Final output : 1

8Q. Edge detection

Edge detection identifies the boundaries of object by detecting sudden changes in pixel intensity. It helps in image segmentation and feature extraction

8(b) Calculation

Sobel Horizontal mask:

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Image matrix

$$\begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$$

Gradient at the center

$$\begin{aligned} & (-1 \times 10) + (-2 \times 10) + (-1 \times 10) + (1 \times 30) + (2 \times 30) + (1 \times 30) \\ & = -10 - 20 - 10 + 30 + 60 + 30 = 80 \end{aligned}$$

Q9. Region Growing

Region growing is a segmentation method that starts from a seed pixel and adds neighboring pixels whose intensity difference is within a specified threshold, forming a homogeneous region.

(b) Calculation

Given pixel: $[48, 50, 52]$

Seed = 50, Threshold difference = 5

$$\bullet |48 - 50| = 2 \leq 5$$

$$\bullet |50 - 50| = 0 \leq 5$$

$$\bullet |52 - 50| = 2 \leq 5$$

Final region: $[48, 50, 52]$ (All pixels belong to the region).

Q10. Combined Enhancement & Segmentation

Smoothing is applied before segmentation to reduce noise and unwanted variations. This helps achieve more accurate and reliable segmentation results.

(b) Calculation

Given Image Values: $[45, 48, 51]$

Apply threshold ($T = 50$):

- $48 < 50 \rightarrow 0$

final segmented output: 0

Q11. Gamma correction

Gamma correction adjusts image brightness, when $\gamma < 1$, the image becomes brighter dark pixels.

(b) justification

use $s = c r^\gamma$ with $c = 1$ & $\gamma = 0.5$ because it brightens low-intensity pixels and improves visibility.

(c) & (d)

Given: $\gamma = [25, 100, 225]$

$$s = \sqrt{r}$$

- $\sqrt{25} = 5$

- $\sqrt{100} = 10$

- $\sqrt{225} = 15$

Final output: $[5, 10, 15]$

q12 (a): Contrast Stretching

Contrast stretching improves image contrast by expanding the intensity range, making details more visible

(b) Justification

A linear transformation maps the intensity range $[50, 150]$ to $[0, 255]$, utilizing the full grayscale range.

(c) & (d)

$$S = \frac{(I - 50) \times 255}{150 - 50}$$

• $I = 50 : S = 0$

• $I = 100 : \frac{50 \times 255}{100} = 127.5 \approx 128$

• $I = 150 : S = 255$

Final output : $[0, 128, 255]$

q13 (a) Histogram Equalization

Histogram equalization improves image contrast by redistributing pixel intensities, making details more visible

(b) Justification

The cumulative distribution function (CDF) is used to map old

3(a). Cumulative distribution

Intensity	frequency	Cumulative
0	4	4
1	6	10
2	10	20
3	6	26

(d) New intensity levels

using $L = 16$ (4-bit image)

• $0 \rightarrow 2$

• $1 \rightarrow 6$

• $2 \rightarrow 12$

• $3 \rightarrow 15$

(e) Interpretation

The pixel intensities are spread over a wider range
improving image contrast & making objects easier to distinguish

q15. median filtering

median filtering remove salt & pepper noise while

Preserving image edges

(b) Justification

The median filter is preferred because it effectively removes impulse noise without blurring the image.

15 (c) & (d) Calculation

$$\begin{bmatrix} 10 & 200 & 10 \\ 200 & 50 & 200 \\ 10 & 200 & 10 \end{bmatrix}$$

Arrange the values in ascending order

10, 10, 10, 10, 50, 200, 200, 200, 200

median (center value) = 50

final output : 50

15(e) Interpretation

The median filter removes salt and pepper noise while preserving important image details and edges.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation